Adaptive Wavelet-Based Image Compression Using Bayesian Thresholding and Run-Length Encoding

Prof. S Kalyani Neer Malik Piyush Puru Piyush Som Vidharbh
Department of Computer Science and Engineering
VIT-AP University, India

***Abstract*—This paper presents an efficient image compression framework combining wavelet transform, Bayesian thresholding, adaptive quantization, and run-length encoding. The method reduces redundancy while preserving visual quality and achieves a balance between compression ratio and reconstruction accuracy.**

***Index Terms*—Image compression, wavelet transform, Bayesian thresholding, quantization, run-length encoding**

I. Introduction

The rapid growth of digital images demands efficient compression techniques. Image compression reduces storage requirements while maintaining visual quality.

Earlier work such as [1] explored the use of self-organizing feature maps with wavelet transforms. While effective, such approaches involve higher computational complexity.

This work proposes a simpler hybrid model combining wavelet transform with statistical thresholding and encoding.

II. Literature Review

Wavelet-based methods enable multi-resolution representation of images. Thresholding improves sparsity, while encoding reduces redundancy. This work integrates these concepts into a unified framework.

III. Theoretical Background

A. Discrete Wavelet Transform

$$I(x,y) = \sum c_{i,j}\psi_{i,j}(x,y) \tag{1}$$

B. Energy Distribution

$$E_{LL} \gg E_{LH}, E_{HL}, E_{HH} \tag{2}$$

IV. System Architecture

A. Compression Pipeline

The compression pipeline systematically reduces image redundancy through a sequence of transformations. Initially, the discrete wavelet transform decomposes the image into multiple frequency subbands, enabling efficient representation of both spatial and frequency information. The majority of the signal energy is concentrated in the low-frequency components, while high-frequency components capture edges and fine details.

To enhance sparsity, Bayesian thresholding is applied to eliminate insignificant coefficients that contribute minimally to visual perception. This step significantly reduces data size without introducing major distortion. Following this, adaptive quantization adjusts the precision of coefficients based on local variance, ensuring that visually important regions retain higher accuracy while less significant regions are compressed more aggressively.

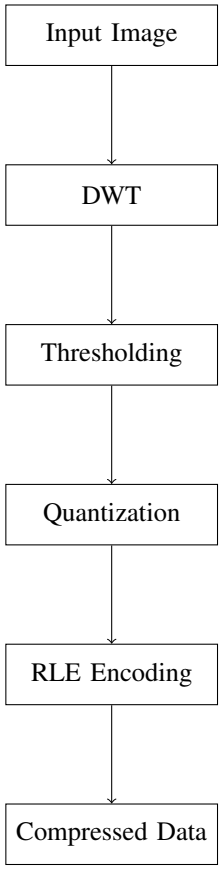


Fig. 1. Compression Pipeline

Finally, run-length encoding exploits consecutive repeated values in the quantized data, further minimizing storage requirements. This multi-stage process ensures an effective trade-off between compression ratio and reconstructed image quality.

The compression process consists of wavelet decomposition, thresholding, quantization, and encoding as shown in Fig. 1.

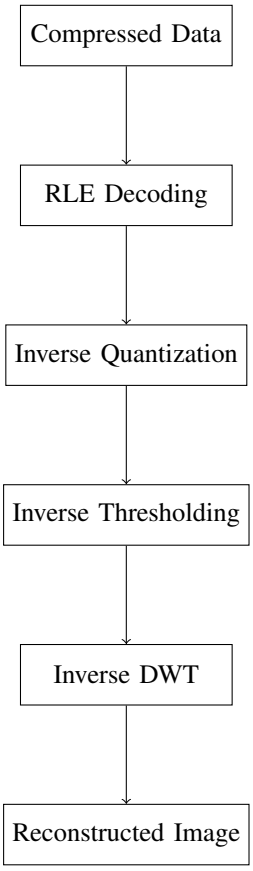


Fig. 2. Reconstruction Pipeline

B. Reconstruction Pipeline

The reconstruction process reverses compression steps including decoding, inverse quantization, and inverse wavelet transform as shown in Fig. 2.

V. Methodology

A. Wavelet Decomposition

$$I \rightarrow LL + LH + HL + HH \tag{3}$$

B. Bayesian Thresholding

$$T = \frac{\sigma_n^2}{\sigma_x} \tag{4}$$

$$\hat{w} = \text{sign}(w)\max(|w| - T, 0) \tag{5}$$

C. Adaptive Quantization

$$Q = Q_0(1 + Var(w)) \tag{6}$$

$$q = round(w/Q) \tag{7}$$

VI. Performance Metrics

$$MSE = \frac{1}{MN}\sum(I - \hat{I})^2 \tag{8}$$

$$PSNR = 10\log_{10}\frac{255^2}{MSE} \tag{9}$$

$$CR = \frac{Original}{Compressed} \tag{10}$$

VII. Experimental Results

A. Cameraman

TABLE I
Cameraman Results

Size	Level	PSNR	CR	RMSE
256	L3	34.16	11.14	0.019
128	L3	33.30	10.06	0.021
64	L1	30.19	10.05	0.030

VIII. Graph Analysis

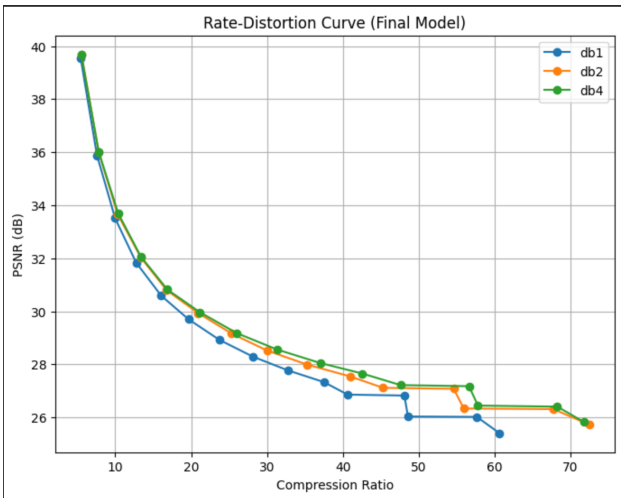


Fig. 3. PSNR vs Compression Ratio (Cameraman)

IX. Conclusion

The proposed method efficiently compresses images while maintaining visual quality using wavelet transform and statistical processing.

X. Future Work

- Deep learning integration
- Real-time compression systems
- Color image extension

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